

Comparative Analysis of Neuroinflammation-Associated Gene Expression in TCGA Lower-Grade Glioma and Glioblastoma

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Abstract

Background: Neuroinflammation-associated genes may exhibit different expression patterns across glioma tumour groups. This exploratory study compared the expression values of ten selected genes between The Cancer Genome Atlas (TCGA) lower-grade glioma (LGG) and glioblastoma (GBM) primary tumour samples.

Methods: Gene-expression data were obtained from the UCSC Xena TCGA lower-grade glioma and glioblastoma dataset ([TCGA.GBMLGG.sampleMap/HiSeqV2](#)) and matched with clinical annotations. After restricting the analysis to primary tumour samples, 516 LGG and 154 GBM samples were included. Expression distributions were compared using two-sided Mann-Whitney U tests, with p-values adjusted across the ten gene-level comparisons using the Benjamini-Hochberg procedure. Cliff's delta was calculated with positive values indicating a tendency toward higher expression values in GBM.

Results: Seven genes showed statistically significant differences in expression distributions after multiple-testing correction. CXCL10, CASP1, PYCARD, IL18, AIF1 and IL1B had higher median expression values in GBM than in LGG, whereas TLR4 had a lower median. TNF, NLRP3 and CSF1R did not meet the adjusted significance threshold. CXCL10 showed the largest observed median difference (GBM minus LGG = 4.3913; adjusted $p = 3.78 \times 10^{-50}$; Cliff's $\delta = 0.798$).

Conclusions: This exploratory analysis identified differences in dataset-provided expression values for several selected genes between LGG and GBM primary tumour samples. These exploratory associations do not establish causation, identify the cellular source of expression signals, demonstrate longitudinal tumour progression, or account for underlying molecular subtype composition (e.g., IDH mutation status).

Keywords

Glioma; glioblastoma; lower-grade glioma; neuroinflammation; gene expression; TCGA; UCSC Xena

1. Introduction

Gliomas are a diverse group of tumours of the central nervous system. Lower-grade glioma (LGG) and glioblastoma (GBM) represent clinically and biologically distinct tumour groups. Large-scale molecular studies, including The Cancer Genome Atlas (TCGA), have helped characterize the genomic and molecular diversity of gliomas (Cancer Genome Atlas Research Network, 2015).

Inflammatory signalling and immune-associated processes are relevant areas of investigation in tumour biology. Genes involved in inflammasome-related signalling, cytokine pathways, innate immune responses and myeloid-cell-associated biology may show different expression patterns across tumour samples. However, bulk-tumour expression measurements reflect a mixture of cell types and cannot, by

themselves, establish which cells produced a given transcript or whether expression differences have a causal role in tumour biology.

The present study performs an exploratory comparison of ten selected neuroinflammation-associated genes, NLRP3, PYCARD, CASP1, IL1B, IL18, TLR4, TNF, CXCL10, AIF1 and CSF1R, between TCGA LGG and GBM primary tumour samples. The research question was: How do expression patterns of selected neuroinflammation-associated genes differ between TCGA lower-grade glioma and glioblastoma?

2. Materials and Methods

2.1 Data source and sample selection

Gene-expression data were obtained from the UCSC Xena platform using the TCGA lower-grade glioma and glioblastoma (GBMLGG) dataset, identified as [TCGA.GBMLGG.sampleMap/HiSeqV2](#) (UCSC Xena Accessed on Oct 4th, 2026). The dataset metadata describes the data type as RNA-seq gene expression and the expression unit as $\log_2(\text{norm_count} + 1)$. Clinical annotations were used to identify tumour type and sample type. The analysis was restricted to primary tumour samples classified as LGG or GBM. After filtering and matching samples between the expression and clinical datasets, 516 LGG samples and 154 GBM samples were included (670 samples total). The dataset contains 702 samples overall, which is distinct from the final analysis set of 670 primary tumour samples.

The expression values were used as provided in the downloaded dataset, whose metadata report the unit as $\log_2(\text{norm_count} + 1)$. The upstream normalization procedure was not independently validated as part of this analysis.

2.2 Candidate genes

Candidate genes were selected based on established roles in distinct but complementary aspects of neuroinflammatory and innate immune biology in glioma. NLRP3, PYCARD, CASP1, IL1B, and IL18 represent core components of inflammasome assembly and downstream cytokine output; inflammasome signaling has been implicated in both tumor-promoting and tumor-suppressive roles across cancer types, including glioma (Hamarsheh & Zeiser, 2020). CXCL10 was included as a chemokine previously shown to increase with glioma grade and to be associated with tumor cell proliferation (Maru et al., 2008). AIF1 and CSF1R were selected as established markers of microglia and tumor-associated macrophages, which constitute a major component of the glioma immune microenvironment and whose abundance correlates with tumor grade (Wang et al., 2025). TLR4 was included given its established, though context-dependent, role in glioma innate immune signaling, including evidence of reduced TLR4 expression in glioblastoma stem cells as a mechanism of immune evasion (Alvarado et al., 2017). TNF was included as a broadly studied pro-inflammatory cytokine relevant to the tumor microenvironment across cancer types. The panel was assembled by the authors prior to data analysis based on this literature and was not data-driven. Collectively, the selected genes were intended to represent several complementary components of neuroinflammatory and innate immune biology rather than to comprehensively characterize the neuroinflammatory transcriptome.

2.3 Statistical analysis

For each candidate gene, the median expression value was calculated separately for LGG and GBM, and the median difference was defined as the GBM median minus the LGG median. Expression distributions

were compared using two-sided Mann-Whitney U tests with the asymptotic method. The Mann-Whitney U statistic is reported alongside p-values and effect sizes for each gene (Table 1). The resulting p-values were adjusted across the ten gene-level comparisons using the Benjamini-Hochberg procedure. An adjusted p-value below 0.05 was considered statistically significant.

Cliff's delta was calculated with the direction GBM minus LGG; positive values indicate a tendency toward higher expression values in GBM, whereas negative values indicate a tendency toward higher values in LGG. The Mann-Whitney U test assesses differences between distributions and is not, without additional assumptions, a test of medians alone. Statistical analyses were performed in Python 3.13.15 using pandas 2.2.3, NumPy 2.1.3, SciPy 1.16.3 for the Mann-Whitney U tests, and statsmodels 0.15.0 for Benjamini-Hochberg correction. The primary inferential comparison concerned the distributions of expression values between groups; medians and median differences were used as descriptive summaries.

2.4 Independent cohort validation

Independent validation was performed using the Chinese Glioma Genome Atlas (CGGA) mRNA-seq 693 cohort. Clinical and expression data were matched by sample identifier, and the analysis was restricted to primary tumour samples. Samples classified as WHO grade II or III were grouped as LGG, while WHO grade IV samples were grouped as GBM, yielding 282 LGG and 140 GBM primary tumour samples. The same ten candidate genes and statistical framework used for the TCGA analysis were applied to the CGGA cohort. CGGA expression values were obtained from the cohort's STAR+RSEM FPKM expression matrix and therefore were not directly compared in absolute magnitude with the TCGA dataset-provided expression values. Validation was assessed by comparing the direction and statistical significance of gene-level differences between the two cohorts. P-values from the ten CGGA gene-level comparisons were adjusted using the Benjamini-Hochberg procedure.

2.5 Clinical covariate sensitivity analysis

A secondary sensitivity analysis was performed in the TCGA cohort to assess whether the observed between-group differences were robust to adjustment for age at initial pathological diagnosis and gender. For each candidate gene, an ordinary least-squares regression model was fitted with dataset-provided gene-expression value as the dependent variable and tumour group, age, and gender as predictors. The tumour-group coefficient represents the adjusted difference between GBM and LGG. Analyses were restricted to samples with complete information for tumour group, age, and gender, resulting in 668 complete cases. P-values for the GBM group coefficient were adjusted across the ten genes using the Benjamini-Hochberg procedure. This regression analysis was treated as a secondary sensitivity analysis, while the two-sided Mann-Whitney U analysis remained the primary comparison.

3. Results

3.1 Sample composition

Following sample-type filtering and matching to clinical annotations, the TCGA analysis included 516 LGG and 154 GBM primary tumour samples (670 total). Recurrent tumour and solid tissue normal samples were excluded. The groups also differed in age distribution, with median ages of 41 years for LGG and 60 years for GBM among samples with available age information. Gender was included in the secondary sensitivity analysis.

3.2 Primary TCGA analysis

Seven of the ten candidate genes showed significant differences in expression-value distributions between LGG and GBM after Benjamini-Hochberg correction. CXCL10 showed the largest observed median difference, with higher values in GBM, followed by CASP1, PYCARD, IL18, AIF1 and IL1B. TLR4 showed lower values in GBM. TNF, NLRP3 and CSF1R did not reach the adjusted significance threshold.

The direction and magnitude of these distributional differences are summarized by Cliff's delta in Table 1. Median differences are presented as descriptive summaries rather than as the estimand of the Mann-Whitney U test.

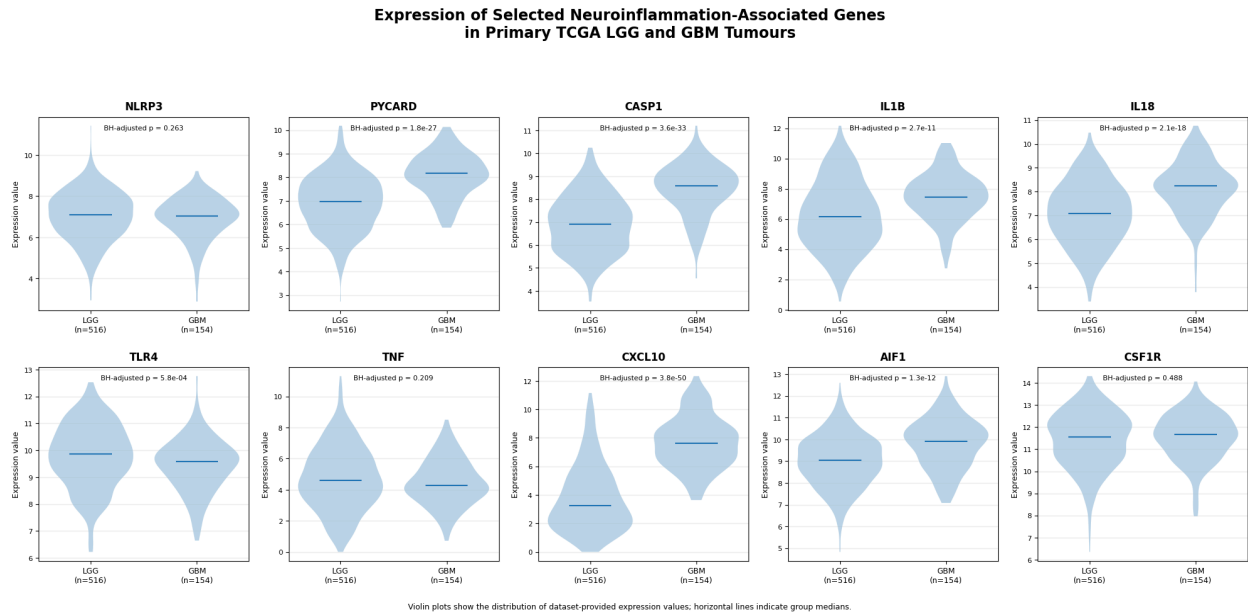


Figure 1. Expression of neuroinflammation-associated genes in primary TCGA lower-grade glioma (LGG) and glioblastoma (GBM) tumours. Violin plots compare expression values for ten candidate genes between LGG ($n = 516$) and GBM ($n = 154$) primary tumour samples. Horizontal lines indicate group medians. Adjusted p -values are from two-sided Mann-Whitney U tests with Benjamini-Hochberg correction across ten gene-level comparisons. The UCSC Xena dataset metadata reports expression values in $\log_2(\text{norm_count} + 1)$ units.

Table 1. Comparison of neuroinflammation-associated gene expression between LGG and GBM.

Gene	LGG median	GBM median	Median diff*	U statistic	Cliff's δ	Raw p-value	Adjusted p-value
CXCL10	3.243	7.635	4.391	8020	0.798	3.78×10^{-51}	3.78×10^{-50}
CASP1	6.903	8.596	1.693	14159	0.644	7.18×10^{-34}	3.59×10^{-33}
PYCARD	6.967	8.183	1.217	16613	0.582	5.48×10^{-28}	1.83×10^{-27}
IL18	7.088	8.250	1.163	21068.5	0.470	8.47×10^{-19}	2.12×10^{-18}

AIF1	9.033	9.904	0.871	24570	0.382	6.36×10^{-13}	1.27×10^{-12}
IL1B	6.142	7.441	1.299	25533	0.357	1.63×10^{-11}	2.72×10^{-11}
TLR4	9.859	9.571	-0.288	47186.5	-0.188	4.06×10^{-4}	5.80×10^{-4}
TNF	4.616	4.283	-0.333	42642.5	-0.073	0.167	0.209
NLRP3	7.090	7.040	-0.051	42228.5	-0.063	0.236	0.263
CSF1R	11.563	11.663	0.100	38270	0.037	0.488	0.488

*Median difference = GBM median - LGG median. Cliff's δ is also oriented as GBM minus LGG; positive values indicate a tendency toward higher expression in GBM. P-values are two-sided Mann-Whitney U tests; adjusted p-values use the Benjamini-Hochberg procedure across ten genes. Values are rounded for presentation.

3.3 Independent CGGA validation

Independent analysis of the CGGA mRNA-seq 693 cohort included 282 LGG and 140 GBM primary tumour samples. CXCL10, CASP1 and IL1B showed significant differences in the same direction in both TCGA and CGGA after multiple-testing correction, providing the strongest independent support for these findings.

PYCARD, IL18, AIF1 and TLR4 showed the same direction in both cohorts but were not significant after correction in CGGA. TNF and NLRP3 were non-significant in both cohorts, while CSF1R showed opposite directions between cohorts. Because TCGA and CGGA use different processed expression scales, validation was based on within-cohort direction, effect size and statistical significance rather than direct comparison of absolute expression values.

Table 2. Comparison of TCGA and CGGA findings

Gene	TCGA median diff.	TCGA Cliff's δ	TCGA adjusted p	CGGA median diff.	CGGA Cliff's δ	CGGA adjusted p	Validation result
CSF1R	0.0998	0.0368	0.488099	-0.420	-0.0639	0.316994	Opposite direction
TNF	-0.3330	-0.0733	0.209290	-0.015	-0.1135	0.115390	Same direction, neither significant

NLRP3	-0.0508	-0.0628	0.262637	-0.090	-0.0662	0.316994	Same direction, neither significant
PYCAR D	1.2167	0.5819	1.82747×10^{-27}	3.635	0.0717	0.316994	Same direction, not significant in CGGA
IL18	1.1626	0.4697	2.11629×10^{-18}	2.055	0.1008	0.152981	Same direction, not significant in CGGA
AIF1	0.8706	0.3816	1.27131×10^{-12}	8.515	0.0300	0.616088	Same direction, not significant in CGGA
TLR4	-0.2881	-0.1876	5.79954×10^{-4}	-0.220	-0.1172	0.115390	Same direction, not significant in CGGA
CXCL10	4.3913	0.7981	3.78461×10^{-50}	1.195	0.4539	3.02322×10^{-13}	Significant replication
CASP1	1.6927	0.6436	3.59250×10^{-33}	2.540	0.2123	0.00127131	Significant replication
IL1B	1.2995	0.3574	2.71791×10^{-11}	1.325	0.2612	6.19324×10^{-5}	Significant replication

3.4 Age and gender-adjusted sensitivity analysis

A secondary sensitivity analysis evaluated whether the TCGA group differences remained after adjustment for age at initial pathological diagnosis and gender. Complete-case analysis included 668

samples. After Benjamini-Hochberg correction across the ten gene-level group coefficients, seven genes remained significantly associated with tumour group: CXCL10, CASP1, PYCARD, IL18, AIF1, IL1B and CSF1R.

The adjusted GBM coefficients were 3.47284 for CXCL10, 1.46517 for CASP1, 1.23018 for PYCARD, 1.09810 for IL18, 0.965415 for AIF1, 1.40015 for IL1B and 0.384415 for CSF1R. TLR4 was no longer statistically significant after adjustment (adjusted $p = 0.307394$), while NLRP3 and TNF remained non-significant.

The adjusted analysis therefore supported the robustness of several major TCGA findings, particularly CXCL10, CASP1, PYCARD, IL18, AIF1 and IL1B. However, the CSF1R result differed from the primary TCGA analysis and from the direction observed in CGGA, indicating that its association was sensitive to analytical adjustment and should not be interpreted as a robust cross-cohort finding.

Table 3. Age + gender adjusted sensitivity

Gene	N	Adjusted GBM coefficient	Adjusted p	Age coefficient	Male coefficient	BH-adjusted p
CXCL10	668	3.473	<0.001	0.0269	0.135	7.85×10^{-44}
CASP1	668	1.465	<0.001	0.00321	0.0606	1.33×10^{-27}
PYCARD	668	1.230	<0.001	-0.00202	0.0461	7.66×10^{-24}
IL18	668	1.098	<0.001	-0.00146	0.0534	2.45×10^{-15}
AIF1	668	0.965	<0.001	-0.00948	0.0638	2.60×10^{-14}
IL1B	668	1.400	<0.001	-0.0108	0.149	3.93×10^{-10}
CSF1R	668	0.384	0.00272	-0.0170	0.0120	0.00388

TLR4	668	-0.138	0.246	-0.0128	0.0400	0.307
NLRP3	668	0.0435	0.715	-0.0108	0.0222	0.795
TNF	668	-0.0443	0.825	-0.0152	0.398	0.825

4. Discussion

In this exploratory comparison of TCGA lower-grade glioma (LGG) and glioblastoma (GBM) primary tumour samples, several selected neuroinflammation-associated genes showed differences in dataset-provided expression values between tumour groups. CXCL10 showed the largest observed median difference on the reported expression scale, with higher values in GBM, followed by CASP1, PYCARD, IL18, AIF1 and IL1B. TLR4 showed lower values in GBM, while TNF, NLRP3 and CSF1R did not reach the adjusted significance threshold in the primary analysis. These findings indicate that the selected inflammatory and immune-associated genes do not show a uniform pattern across the two TCGA tumour groups.

The strongest independent support was observed for CXCL10, CASP1 and IL1B, which showed differences in the same direction and remained statistically significant after multiple-testing correction in the independent CGGA cohort. PYCARD, IL18, AIF1 and TLR4 showed the same direction of difference across both cohorts but were not statistically significant in CGGA. TNF and NLRP3 were non-significant in both cohorts, whereas CSF1R showed opposite directions between TCGA and CGGA. Because the two datasets use different processed expression scales, these results should be interpreted in terms of within-cohort direction, effect size and statistical significance rather than direct comparison of absolute expression values. Overall, the independent analysis provides partial validation of the primary findings, with the most consistent evidence observed for CXCL10, CASP1 and IL1B.

The biological interpretation of these findings requires caution. CXCL10 has previously been associated with glioma biology and tumour cell proliferation, while TLR4 has been reported to have context-dependent roles in glioblastoma, including reduced expression in glioblastoma stem cells. The present analysis cannot determine whether the observed expression patterns reflect tumour-cell-intrinsic biology, changes in immune or stromal cell abundance, or other differences in tumour composition. In particular, bulk-tumour expression measurements combine transcripts from multiple cell populations and therefore cannot establish the cellular source of an observed signal. Similarly, increased transcript abundance should not be interpreted as direct evidence of increased pathway activity or functional immune activation.

The age- and gender-adjusted sensitivity analysis supported the robustness of several primary TCGA associations. CXCL10, CASP1, PYCARD, IL18, AIF1 and IL1B remained significantly associated with tumour group after adjustment for these demographic covariates. TLR4 was no longer significant after

adjustment, suggesting greater sensitivity to covariate specification. CSF1R became significant in the adjusted model despite being non-significant in the primary analysis and showing an opposite direction in CGGA. This discrepancy indicates that the CSF1R association is sensitive to analytical specification and should not be considered a robust cross-cohort finding.

Several limitations should be considered when interpreting these results. First, the LGG and GBM groups represent historical TCGA classifications rather than the molecularly defined entities used in the current WHO classification. Differences in molecular composition, including IDH status and other genomic features, may therefore contribute to the observed expression differences. Second, the analysis was cross-sectional and did not follow individual tumours over time; consequently, the results cannot establish progression from LGG to GBM. Third, the study examined a focused ten-gene panel selected to represent several components of neuroinflammatory and innate immune biology rather than a comprehensive transcriptomic characterization of neuroinflammation. Fourth, the primary analysis was not adjusted for molecular covariates or treatment history, and the available clinical sensitivity analysis was limited to age and gender. Finally, the analysis used dataset-provided expression values and did not independently validate the upstream normalization procedure.

Taken together, the results provide exploratory evidence that selected neuroinflammation-associated genes show different expression patterns between TCGA LGG and GBM primary tumours, with CXCL10, CASP1 and IL1B showing the most consistent support across independent cohorts. Further studies incorporating molecularly defined tumour subgroups, additional clinical covariates and cell-type-resolved or single-cell data will be important for determining the biological basis of these expression patterns.

5. Conclusion

This exploratory analysis identified differences in dataset-provided expression values for several selected neuroinflammation-associated genes between TCGA LGG and GBM primary tumour samples. CXCL10, CASP1 and IL1B showed the strongest and most consistent evidence across the primary TCGA analysis and independent CGGA validation, while PYCARD, IL18, AIF1 and TLR4 showed less consistent independent support. Age- and gender-adjusted analysis supported several major TCGA associations but also demonstrated that some findings, particularly CSF1R and TLR4, were sensitive to analytical specification.

These results represent associations in bulk tumour transcriptomic data and do not establish causation, pathway activation, cellular origin, or longitudinal tumour progression. Further molecularly stratified and cell-type-resolved studies are needed to determine the biological context and robustness of these expression patterns.

Data Availability

The study used publicly available TCGA lower-grade glioma and glioblastoma gene-expression and clinical data accessed through UCSC Xena. The expression dataset was

[TCGA.GBMLGG.sampleMap/HiSeqV2](https://xenabrowser.net/datapages/?dataset=TCGA.GBMLGG.sampleMap/HiSeqV2), with metadata-reported expression units of

$\log_2(\text{norm_count} + 1)$ and dataset version 2017-09-08. The dataset is available at:

<https://xenabrowser.net/datapages/?dataset=TCGA.GBMLGG.sampleMap%2FHiSeqV2&host=https%3A%2F%2Ftcga.xenahubs.net>. Data accessed October 4, 2026.

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